

Project Deliverable H:
Prototype III and Customer Feedback

Group 16: The Sample Snatchers

Eden Irwin, Isaac Pinarski

Radhe Pandey, and Tiam Morrow-Rogers

Faculty of Engineering – University of Ottawa

GNG1103: Engineering Design

Dr. David Knox

March 23, 2025

Table of Contents

1.0 Introduction.....	5
2.0 Prototype V - Movement, Sampling and Power Systems.....	6
2.1 Prototype Description & Justification.....	6
2.2 Prototype Testing.....	10
2.3 Testing Conclusion.....	14
3.0 Prototype VI - User Interface.....	15
3.1 Prototype Description & Justification.....	15
3.2 Prototype Test Plan & Results.....	15
3.3 Prototype Assembly.....	16
4.0 Updates to Specifications and Design.....	17
4.1 Bill of Materials (BOM).....	17
4.2 Changes to Target Design Specifications.....	19
5.0 Conclusion.....	20
5.1 Results & Conclusion.....	20
5.2 Task Plan Update.....	20

List of Tables

Table 4.1: Updated Bill of Materials.....	17
Table 4.3-1: EDS - Functional Requirements.....	19
Table 4.3-2: EDS - Constraints.....	19
Table 4.3-3: EDS - Non-Functional Requirements.....	19

List of Figures

Figure 2.1-1: Prototype V (Side View).....	6
Figure 2.1-2: Prototype V (Top View).....	7
Figure 2.1-3: Motor Mount Prototypes.....	7
Figure 2.1-4: Additional Motor Mount Prototypes.....	8
Figure 2.1-5: Sampling Bit (Idle Configuration).....	8
Figure 2.1-6: Sampling Bit (Sampling Configuration).....	9
Figure 2.1-7: Battery Pack Assembly (Front View).....	9
Figure 2.1-8: Battery Pack Assembly (Side View).....	10
Figure 2.2-1: Battery Tester Assembly.....	10
Figure 2.2-2: Battery Pack Assembly (Front View).....	11
Figure 2.2-3: Battery Pack Assembly (Top View).....	11
Figure 2.2-4: TPU Rings.....	12
Figure 2.2-5: Rubber Bands.....	12
Figure 2.2-6: Sampling Head (Front View).....	13
Figure 2.2-7: Sample Head with Tooling.....	13
Figure 3.3-1: UX WireFrame (Google Slides).....	16
Figure 3.3-2: Application Development (Figma).....	16

Abstract

This report presents the development and evaluation of prototypes for the third phase of the prototyping and testing stage of the Design Thinking process. Two main prototypes have been developed and tested: a comprehensive high-fidelity prototype integrating movement, sampling, and power systems (Prototype V) and the user interface (Prototype VI). The corresponding test and task plans have been updated for each prototype while justifying reasoning and content. The purpose of these prototypes expand on the desired objectives by referencing previous work and applying prior knowledge and skills to incorporate into the current comprehensive prototype.

The comprehensive, high-fidelity physical prototype (Prototype V) includes an assessment of power efficiency, movement precision, and sampling effectiveness. Unfortunately, the sampling mechanism faced challenges in collecting adequate material. Meanwhile, the user interface (Prototype VI) was successfully developed and tested, receiving positive feedback on usability and design.

In brief, this document details the progression made towards the development of the design through two prototypes while facing obstacles in testing but expanding our knowledge and procedure for future work. The team will continue to work effectively in preparation for Design Day on March 27, 2025.

1.0 Introduction

This report details one of the final prototyping and testing phases, which strives to evaluate the movement and sampling in a fully developed comprehensive system. Previous focused prototypes of these separate subsystems have been modelled, tested, and implemented in this design. An additional user interface prototype has been developed to provide an interactive experience for the user and portray the device's operation process.

This document aims to display the detailed development process of the comprehensive design and user interface to achieve the previously defined objectives the team has set out in the test plan to describe the critical analytical guiding questions directly. The team will evaluate the targeted goals, set with specific tests and measurable units through an extensive prototyping test plan to exhibit a critical analysis and results. The team proceeds to gather feedback on the prototypes to deepen our understanding and evaluate the feasibility and functionality of the design. As the testing of the prototype has been fulfilled, the necessary adjustments have been made to the target design specifications and BOM.

This report concludes the third documented prototyping phase. It justifies and explains the reasoning behind the comprehensive prototype and briefly explains the results obtained from previous prototypes. This prototype clearly demonstrates a progression towards the solution's development.

Design Day is March 27, 2025 – The team has updated the Trello task board to complete all required tasks and deliverables for Design Day. Further task plan updates have been made towards the prototyping task plan to include any critical sub-tasks to complete in time for Design Day.

Overall, this documentation summarizes the prototyping progression, justifies results and improvements, and plans future work to complete to achieve a successful design.

2.0 Prototype V - Movement, Sampling and Power Systems

2.1 Prototype Description & Justification

The following is a comprehensive physical Hi-Fi prototype comprising multiple subsystems, including our power, sampling, and movement systems, as defined in previous prototypes. Notable components include the bearing-mounted wheels that allow the device to fit into the tube with extreme precision, preventing slippage and resisting rotational forces caused by the sampling tool head on the front of the device. Some changes were implemented into the bearings from the previous week to aid with resistance to slippage. In orange, a panel is depicted that allows for easy access to electronics to adapt the design on the go as needed. Additionally, a hook on the back of the device will enable us to recover the device after being sent down the tube with a tether.

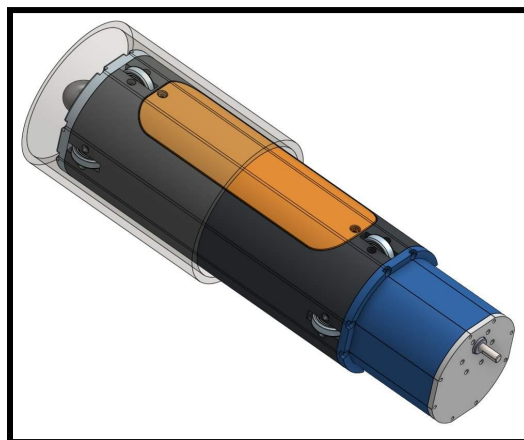


Figure 2.1-1: Prototype V (Side View)



Figure 2.1-2: Prototype V (Top View)

One notable decision made with the tool head is the interchangeable mounting plate for the motor. This allowed us to rapidly test different mounting positions to try to achieve a circumferential overlap between the scraping bits' path and the steel tube wall that would successfully collect a sample. Many iterations were made to mount the motor firmly and with adequate positioning.

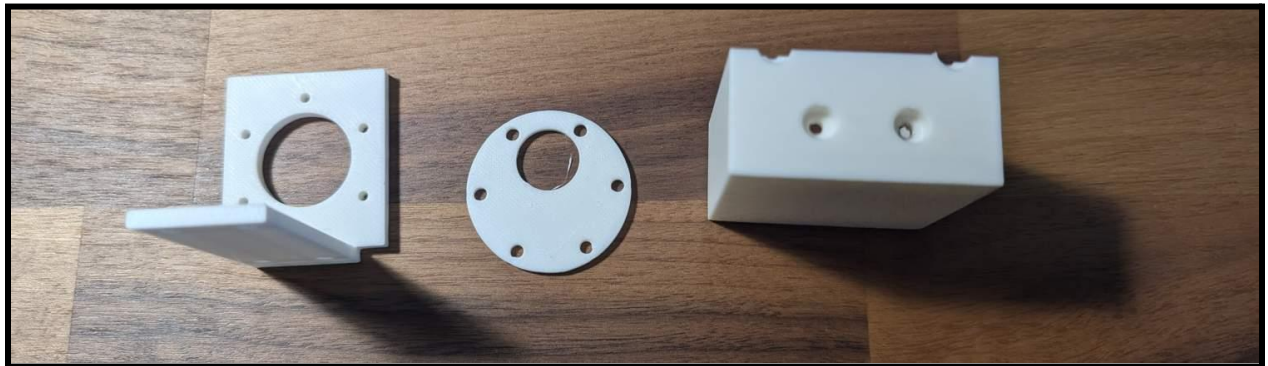


Figure 2.1-3: Motor Mount Prototypes

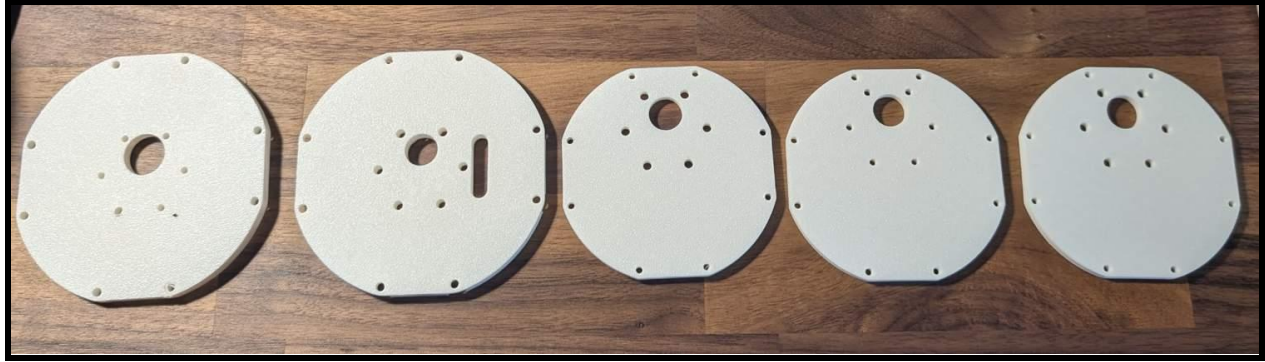


Figure 2.1-4: Additional Motor Mount Prototypes

The scraping tool holder highlighted in orange below implements a grub screw to prevent slippage of the 3D-printed tool holder around the D-mount of the JGB-37 520 motor. This was made to overcome issues encountered the previous week, where our motor tore through the plastic tool holder.

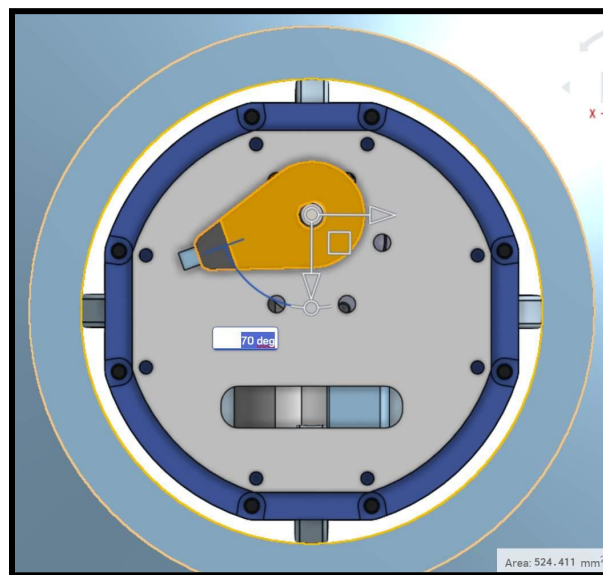


Figure 2.1-5: Sampling Bit (Idle Configuration)

The tool holder was modelled to cut through the tube wall only near the top of its circular path. A hall effect encoder helps determine the motor position along the cutting path, hoping to prevent the tool head from destroying the container directly below.

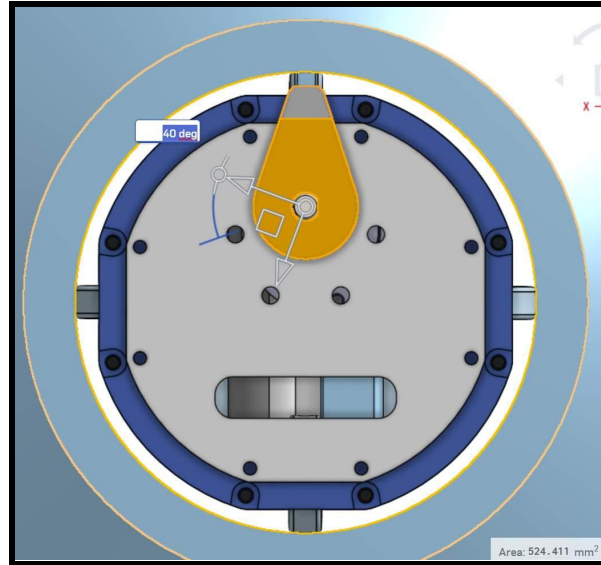


Figure 2.1-6: Sampling Bit (Sampling Configuration)

A custom battery pack was modelled to power the system. This enclosure is version 3 of the design, and it implements mounting points to keep the battery pack stable inside the device. To maintain the 12 volts, we reduced the battery configuration to 3 in series and 1 in parallel. This dropped the amps to 40 from 80 and, most importantly, reduced the pack's weight in half. Our system uses less than 40 amps, so this was not a concern.

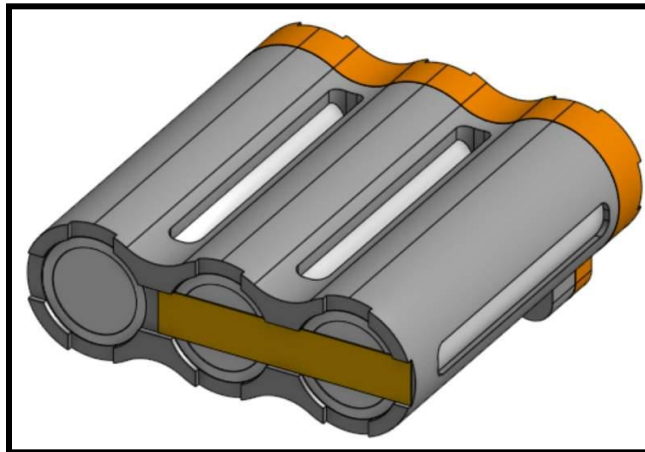


Figure 2.1-7: Battery Pack Assembly (Front View)

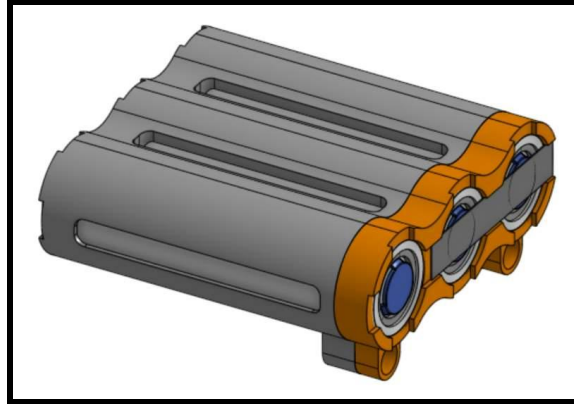


Figure 2.1-8: Battery Pack Assembly (Side View)

2.2 Prototype Testing

Several tests had to be conducted with this prototype, including:

1. Power System Test - Check whether the custom-built battery pack works and powers the design effectively.
2. Movement Test - Check that the bearing-mounted wheels contact and move efficiently and can be pulled by the failsafe system.
3. Scraping Test - Run the motor and check for successfully scraped material.

The batteries for the battery pack were first tested with a battery tester. This ensured that the cells were within safe operational parameters before they were implemented into our design.



Figure 2.2-1: Battery Tester Assembly

The cells could then be assembled into the battery pack enclosure, with nickel strips spot-welded to the terminal ends. These would then be soldered to the battery management system so we could quickly charge and discharge from the battery pack as needed.

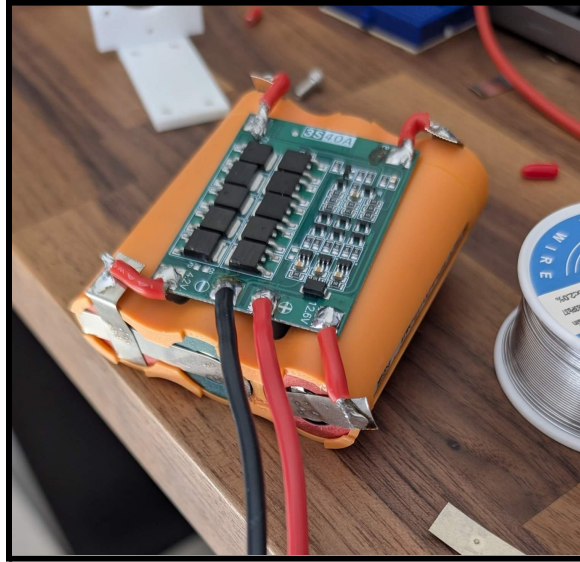


Figure 2.2-2: Battery Pack Assembly (Front View)

With this setup, we charged the pack with a 12V charger, which allowed our cells to reach 95% capacity. This allows for a highly consistent 12-volt discharge for a reasonably long period, as the pack is 4500mAh.

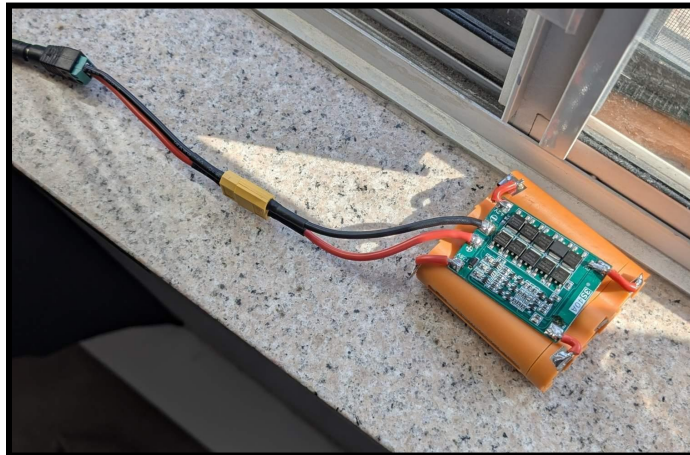


Figure 2.2-3: Battery Pack Assembly (Top View)

Initially, the bearings were mounted to the device and rolled directly on the pipe's inner wall for the movement test. However, we found the bearings to have too low resistance, allowing the device to slide uncontrollably. We then mounted TPU rings to the bearings, hoping the rubber-like material would provide more friction. We still encountered issues, and instead, we found that using the TPU to put rubber bands around the wheel was the most effective solution. The rubber band was compressible, allowing the device to fit in the tube tightly, and the friction seemed sufficient for the scraping motor torque. Several iterations were necessary to fit the wheels and secure the rubber band properly and effectively.



Figure 2.2-4: TPU Rings



Figure 2.2-5: Rubber Bands

At this point, the device fit well, moved efficiently through the pipe, and had the necessary power for all operations. The scraping test was conducted by placing the device into the pipe and

running the motor directly, using the two leads for motor power and connecting them directly to the battery output.

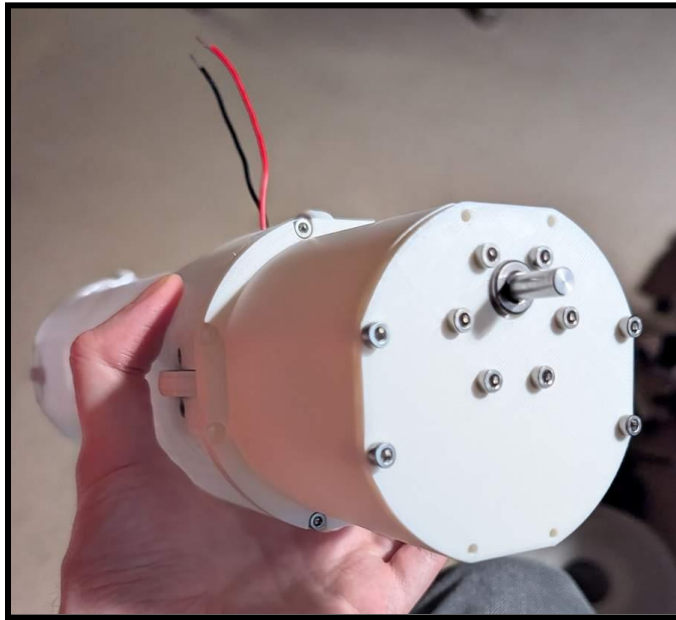


Figure 2.2-6: Sampling Head (Front View)

One version of the tool head included a passthrough window that allowed cables to pass through the front of the device, making repetitive tests easier.

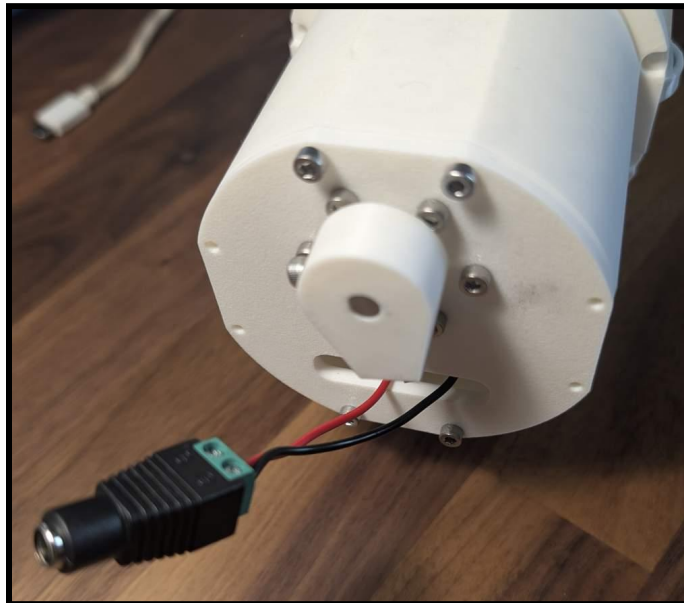


Figure 2.2-7: Sample Head with Tooling

Using this testing setup, we would run the motor and attempt to scrape a sample from the steel pipe. We would then measure the sample collected and determine how many scrapes were necessary to collect an adequate amount.

2.3 Testing Conclusion

Unfortunately, we realized that our manual scraping tests in the previous week were not as practical as we initially surmised. Upon closer inspection, we discovered the scrapes were only removing the oxide layer of the steel pipe. This is also likely the case for every other group. If we ran the motor for a more significant period, silver flakes could be removed. However, the time necessary is too extreme, and scraping deeper was not an option as the motor would overcome the tool bit mount. It doesn't seem like this would improve without metal componentry.

Although our design resulted in unsuccessful sampling, we remain confident it would work with more sufficient components. These include a metal chassis, a 24-volt power supply, a metal armature for the cutting tool, and larger rubberized rollers for the wheels. However, the time and budget to accomplish these are not within the scope of the assigned project, so the group will push forward, focusing on other aspects of the design. We also have concerns over the welding marks of the steel pipe as, for many teams, their scraping method may get caught on the weld and not perform as it would in the real environment.

3.0 Prototype VI - User Interface

3.1 Prototype Description & Justification

This prototype focuses on the user experience as they operate the device. The created app strives for simplicity, aesthetics, and functionality. Although the application is not directly connected to the actual feedback system of the device, the objective of the app allow the user to go through the motions of operating the device from start to finish, such as setting up the required parameters (orientation, distance, mass collected, etc.), starting the device, reviewing the feedback from the device, and reviewing the results of the test.

This additional prototype strives to be original to our solution. It considers the user in other aspects rather than simply handling the device. As the needs of the user/client are met, we wish to exceed the client's expectations to differentiate our team's solution from others.

3.2 Prototype Test Plan & Results

To iterate the prototype test plan, the test objective type is focused on communication and learning/understanding. The app allows the user to understand how to operate the device with a simple, straightforward process. To test the prototype, a visual HiFi-focused prototype has been developed to demonstrate the app's function, where users will be surveyed based on the presented content and ease of navigation. Additional testing was based on the time duration, measured in seconds, that it takes the user to set up and act out the operating process for the device. Further objectives of this prototype are to aim for an app that is easy for the user to follow while ensuring all necessary parameter specifications are met to obtain the required feedback. *(Prototyping Test Plan appended at the end of the document)*

Based on a survey conducted with potential device users, the feedback and comments obtained resulted in a successful prototype. All comments illustrated a visually pleasing UX design, striving for “above and beyond,” meeting all required input fields while also capable of displaying realistic feedback, all without being directly connected to the physically finalized design.

3.3 Prototype Assembly

An initial development of the UX wireframe was displayed in Google Slides to map out the general formatting of the app. Arrows indicate the flow from one frame to another, as Figure 3.3-1 highlights. However, Google Slides does not include an interactive platform for prototyping this display.

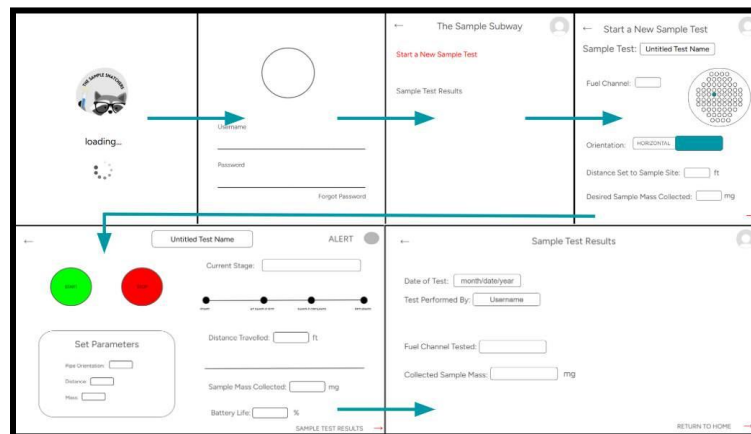


Figure 3.3-1: UX WireFrame (Google Slides)

Further, the prototype to be shown on Design Day will use Figma prototyping to create an interactive user interface that allows the user to actuate the steps the operator would take to use the device. This application's format is oriented around the initial design, but it includes features the user can physically interact with to demonstrate the operating procedure.

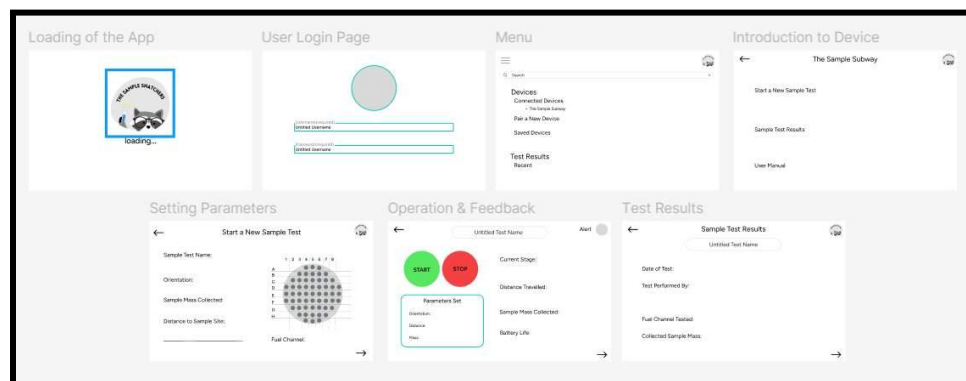


Figure 3.3-2: Application Development (Figma)

4.0 Updates to Specifications and Design

4.1 Bill of Materials (BOM)

The following is the updated BOM. This form of the BOM includes all the parts used throughout the design and remains complete now.

Table 4.1: Updated Bill of Materials

Sample Snatchers - Bill of Materials						
Components that come in bulk quantity have had their price modified to the amount that will realistically be used. This amount will generally be estimated to be higher than lower. Each section has a name scheme to identify components later on quickly. Quantities will be updated more accurately as the final design is developed.						
Battery and Power Delivery (PDXX)						
Item #	Part	Vendor	Description	Qty	Cost	Link for Purchase
PD01	22 awg Wire	Amazon	Wires for connecting components to circuit	1	CA\$0.80	Generic
PD02	XT60 Connectors	Aliexpress	Connectors for battery pack	1	CA\$1.60	Link
PD03	16 awg Wire	Amazon	Wires for 12v components	1	CA\$0.80	Generic
PD04	P42A Molicel Battery 21700	18650 Battery	For battery pack assembly	3	CA\$15.00	Link
PD05	3S Battery Management System	Aliexpress	For safety	1	CA\$2.94	Link
PD07	LM2596	Aliexpress	Step down buck converter, 36V/24V/12V/5V/3V	1	CA\$0.45	Link
Control and Logic (CLXX)						
Item #	Part	Vendor	Description	Qty	Cost	Link for Purchase
CL01	ESP32-WROOM-C3	Elegoo	Controller w/ Wifi + BT	1	CA\$9.99	Link
CL02	H-Bridge Motor Driver BTS7960	Aliexpress	Driving 12V motor for scraping	1	CA\$7.41	Link
CL03	HX711 Sensor	Aliexpress	Strain gauge for mass analysis	1	CA\$1.80	Link

Measurement (MXX)						
Item #	Part	Vendor	Description	Qty	Cost	Link for Purchase
M01	100g Load Cell	Aliexpress	Strain gauge to measure mass	1	CA\$4.93	Link
Structural Components (SCXX)						
Item #	Part	Vendor	Description	Qty	Cost	Link for Purchase
SC01	PLA - ~0.5 KG	Bambu	Filament for chassis	1	CA\$7.99	Link
SC02	M3 Female Threaded Insert	Aliexpress	Heat set threaded inserts for chassis	1	CA\$0.80	Link
SC03	M3/M4 Machine Screws	Aliexpress	Various length M3 Machine Screws	1	CA\$1.00	Generic
SC04	608 ZZ 608RS	Aliexpress	Ball bearings	3	CA\$1.80	Link
Sampling Tooling Components (STXX)						
Item #	Part	Vendor	Description	Qty	Cost	Link for Purchase
ST01	MGEHR1212 + Bits	Aliexpress	Lathe Bit and armature	1	CA\$9.28	Link
ST02	GB37-520 7rpm 12v Motor	Aliexpress	For moving the armature with lathe bit	1	CA\$11.28	Link
Total	CA\$77.87					
Equipment						
	Soldering Iron	Used to connect electrical components together				
	Bambu Lab A1 3D Printer	Fused deposition modelling tool for additive manufacturing of chassis				

4.2 Changes to Target Design Specifications

At this time, our design specifications remain unchanged.

Table 4.3-1: EDS - Functional Requirements

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
	Functional Requirements				
1	Collect a metal sample from the tube wall	=	30-80	mg	Test
2	Completely Removable	=	Yes	N/A	Test
3	Enclosed Sample	=	Yes	N/A	Test
4	Stage Reporting	=	Yes	N/A	Test
5	Visual for process	=	Yes	in, dpi	Test
6	Device speed	<	7	min	Test
7	Power source included	=	Yes	N/A	Analysis
8	Distance Travelled	=	15	ft	Test

Table 4.3-2: EDS - Constraints

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
	Constraints				
1	Weight	<		kg	Analysis
2	Size (Diameter)	<	4	in	Analysis
3	Size (Length)	<	4	ft	Analysis
4	Sample Exposure	<	Yes	N/A	Test
5	Sample Collection	=	30 - 80	mg	Test

Table 4.3-3: EDS - Non-Functional Requirements

Design Specifications		Relation (=, < or >)	Value	Units	Verification Method
	Non-Functional Requirements				
1	Aesthetics	=	Yes	N/A	Survey
2	Safety	=	Yes	N/A	Analysis
3	Corrosion Resistant	=	Yes	N/A	Analysis
4	Operation Time	<	10	mins	Test
5	Creative Solution	=	Yes	N/A	Analysis

5.0 Conclusion

5.1 Results & Conclusion

Throughout the development of Prototype V, the team integrated movement, sampling, and power systems into a comprehensive prototype. However, as previously mentioned, the prototype does not function as the team had hoped. We faced constraints due to the material selection of the prototype. The only way to overcome these constraints would have been to exceed the \$100 budget significantly. However, the user interface prototype was also developed to demonstrate an intuitive and efficient user interaction process.

Despite the design's inconclusive operational ability, the team is still pleased with the product developed. The last step is to mount the container below the motor for design day and prepare the necessary presentation material. An animated video showing it in operation will also be ready to demonstrate the device's capabilities further.

5.2 Task Plan Update

The general task board has been updated in Trello, incorporating task descriptions, estimated durations, additional objectives, team member assignments, and other essential details to support efficient project management. To further illustrate this content, a GANTT chart showing the tasks and timeline has been appended to the submission.

The team task plan has also been updated to include the final sub-tasks as we approach Design Day. These tasks outline specific tasks with notable components such as a detailed description, task time, and designated responsibility. They also acknowledge potential risks and backup plans. As Design Day approaches, these tasks include preparing for the big day while finalizing the design.